25900517 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCMBG200 - GENERAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Give short note on bacterial G+C content. [U] / [CO1]
2. Mention the function of chromophore group and auxochrome group. [K] / [CO2]
3. What are the applications of dry heat sterilization? [K] / [CO3]
4. Name any one anaerobic culture media. [K] / [CO2]
5. What is Bergey's manual? [K] / [CO1]
6. Mention two functions of calcium-dipicolinic acid in endospores. [K] / [CO1]
7. Name two Gram positive cocci. [K] / [CO2]
8. Mention the standard chemical used in the test for comparing disinfectants. [K] / [CO3]
9. Define extracellular polymeric substance (EPS) in bacteria. [K] / [CO1]
10. Mention one advantage of spread plate method. [U] / [CO2]

11. What is periplasmic space? [U] / [CO1]

12. Give a note on bacterial chromosome. [U] / [CO1]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

13. Describe the mechanism of microbial destruction by UV radiations. [K] / [CO3]

14. Explain the fluid mosaic model of bacterial cell membrane. [U] / [CO1]

15. Discuss on non-beta-lactam antibiotics. [U] / [CO3]

16. Differentiate between pili and fimbriae in terms of structure and function. [U] / [CO1]

17. Describe the principle and procedure of endospore staining. [K] / [CO2]

18. Discuss the factors that influence sterilization by heat? [U] / [CO3]

[5x4 = 20]

Part C

Long Essay Questions

Answer any **1** questions

Each question carries **10** marks

19. Write an essay on the composition, structural organisation and function of peptidoglycan. [U] / [CO1]

20. Discuss the applications of lawn culture. Explain stab and stroke culture. [U] / [CO2]

[10x1 = 10]